

Applied Data Science Project

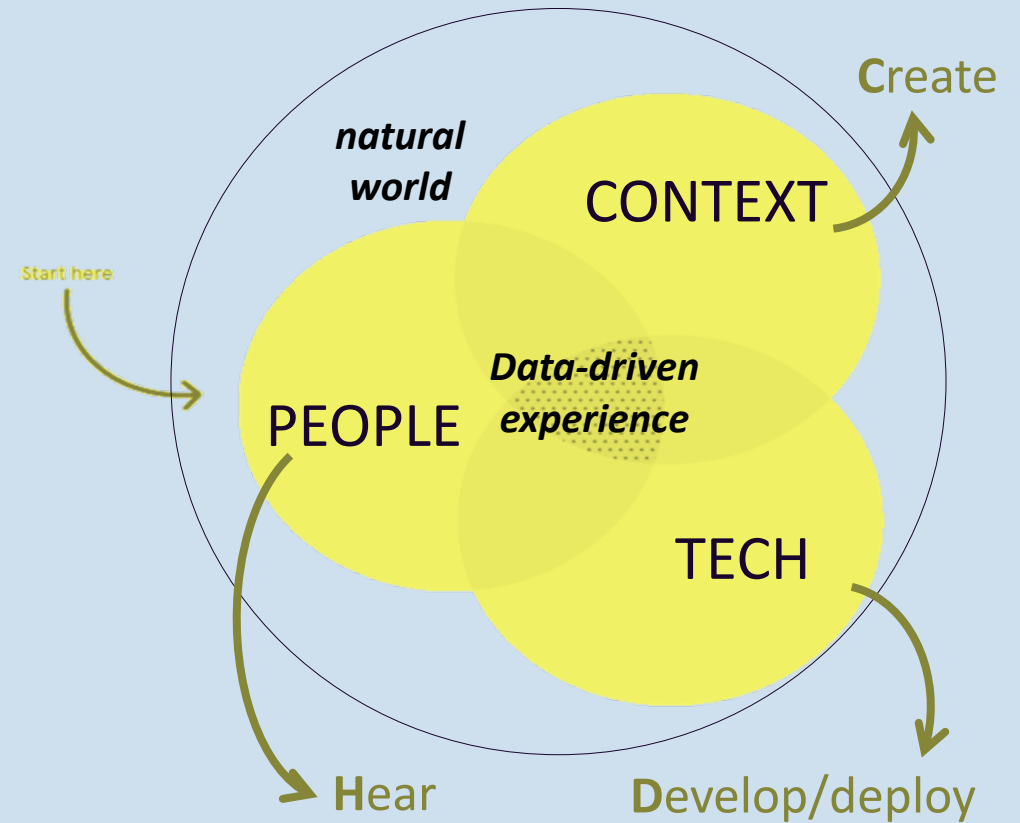
L26 – The stakeholders' mapping Hands-on



**Politecnico
di Torino**

Antonella Frisiello

Multi-perspective thinking



**We are not the user
The user is not the boss...
And not even a colleague, a friend,
the desk mate...**

We need to **access to the users' knowledge**
(both explicit and implicit knowledge).

In many situations, observers, listeners, and researchers are not allowed, so we need for indirect knowledge.
Gathering information to know and map the mental models of users allows us to get out of **our perspectives** and biases and design solutions widely usable and accepted.



Let's make a simple experiment

Considering people currently involved in the training courses of Politecnico di Torino:

- Professors planning and teaching courses
- Administrative staff managing bureaucracy
- Utility staff caring for the accessibility and dignity of places

Let's ask everybody the same question.

What makes a good learning experience for you?

As a student: what makes a good learning experience for you?

Review answers 11

Concrete examples

Good sources

Concept transferring

Understanding why are we learning this or that

Team cohesion

experienced and passionate professor

Interactive and practical learning

Understandable resources

Interaction

30

Appropriate environment

As a professor: what makes a good learning experience?

Review answers 6

Responsive students

Good organization

passion for the profession

Good facilities

Student ask questions

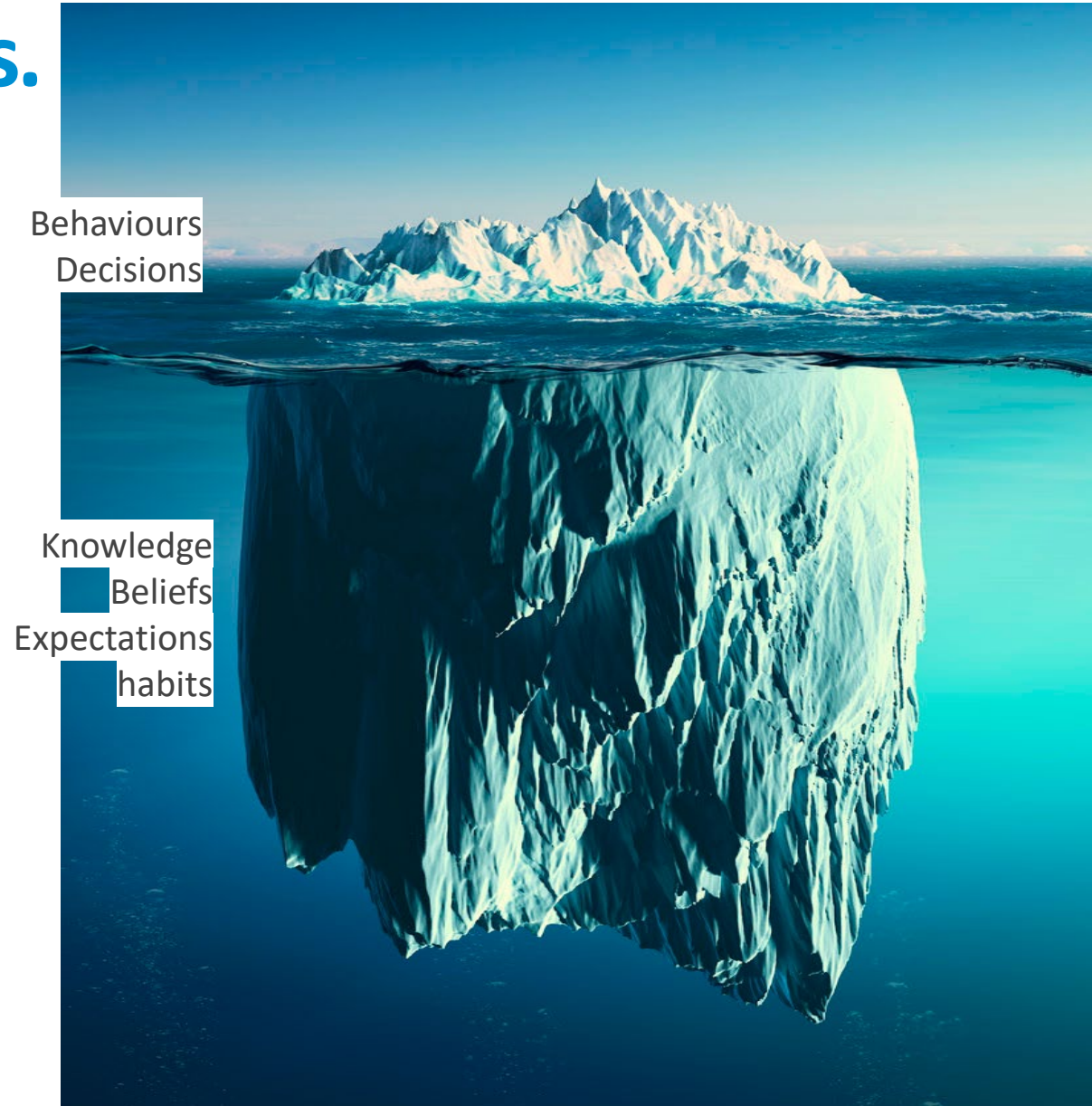
We all have our mental models.

A mental model is a inner **representation of the real perceived world and relations between its various parts.**

A mental model contains knowledge, beliefs, **expectations** and **habits** deriving from our perceptions, **direct and indirect experiences.**

We develop mental models on any aspect of the experience, including the digital systems we use, through intuitive **perceptions of their actions and consequences.**

Mental models drive our decision and behaviours (including the interaction with systems and data).

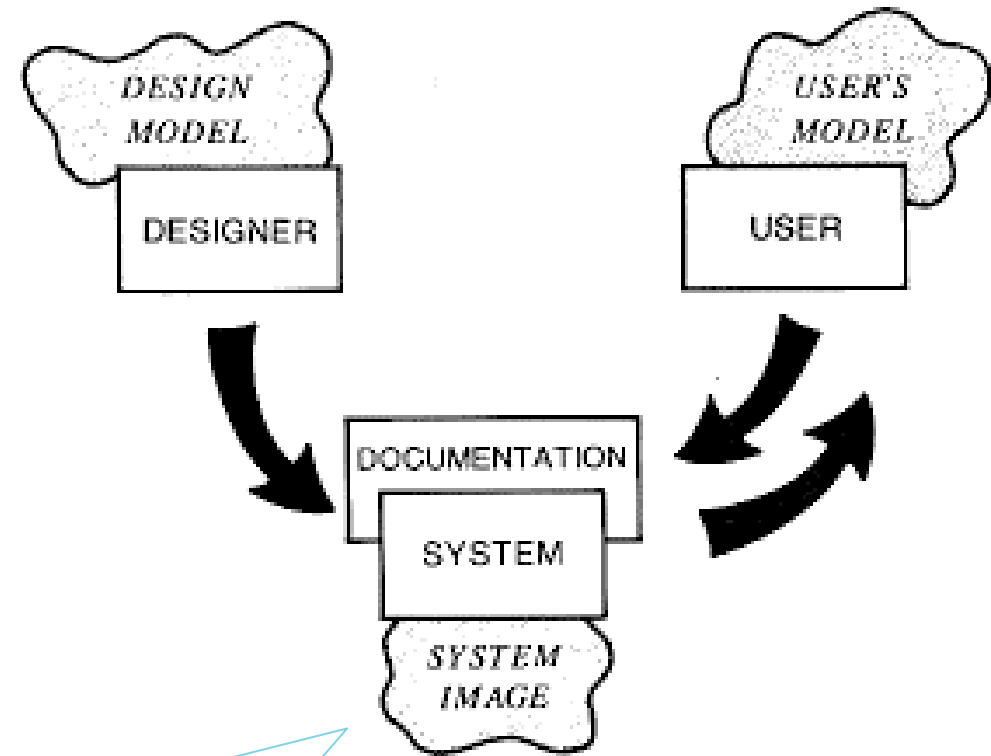


1 project, several mental models

We all create mental models that explain and organise our actions and interactions with the world.

The mental models we create derive from **what we can perceive of a system, its structure and visible behaviours**.

If the system image is incomplete, inadequate, or inconsistent, a weak usage experience will occur.



*The system image is the mental model materialised by design activities, and that will be developed.
It is like a text open to interpretation*

Stakeholders' analysis and mapping



Different people, in different contexts, produce, understand, and use data differently.



*Any person, group, or institution that, positively or negatively, **affects or is affected** by a particular issue or outcome of your project is a stakeholder.*

According to the HCD, stakeholders (people, institutions, or social groups) shall be **involved in the process** **from the early stages and during** the design activities (especially testing and validation)

Sponsors are often those who initiate the project by mobilizing resources and managing activities. Sponsors own the requirement for the project, and if the requirement changes, they must direct the project accordingly.

Which role for clients?

- **Direct or Primary users (CORE players)**
- **Indirect or Secondary users (INVOLVED players)**
- **External stakeholders (INFORMED players)**

Which role for sponsors?

Which role for developers?

1) Stakeholders' analysis

Early identifying who has a stake in the outcome

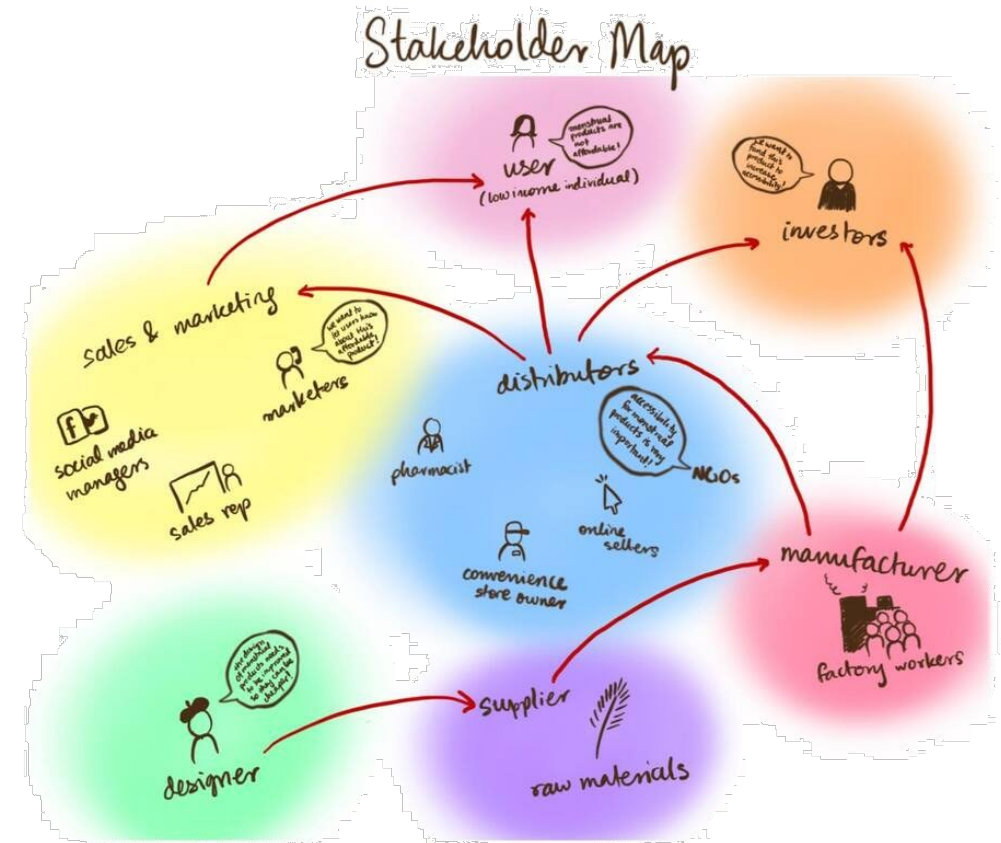
- Directly and indirectly interested in the project

will help steer the project team in the right **direction** and

give a **network** where to find learning opportunities, feedback, support, future chances.

Stakeholder analysis aims at identify:

- **Who** will use the output
- **Who** is investing in the outcome
- **Who** has right of refusal
- **Who** will support the team
- **Who** are the key decision-makers
- **Who** the team should meet with to help get context

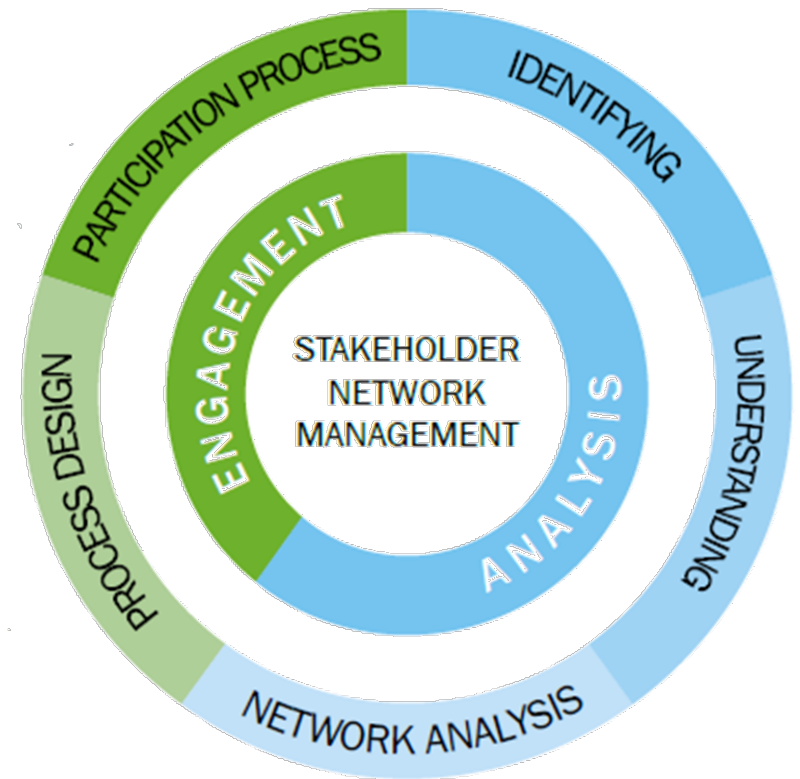


2) Stakeholders' map

Maps are **dynamic supports** to be updated according to the real scenario (spatial/temporal). They enable **different activities**: design, testing, trials, experimentation, distribution and marketing, communication, maintenance...

They provide quick representation of complex information such as:

- **Target users real goals**
- Different **interests**
- **Success** metrics
- **Communication** hints
- Potential **risks**
- **Engagement** of users to be involved in testing activities
- Engagement of **key people** to be informed
- **Negative stakeholders** as well as their adverse effects on the project....

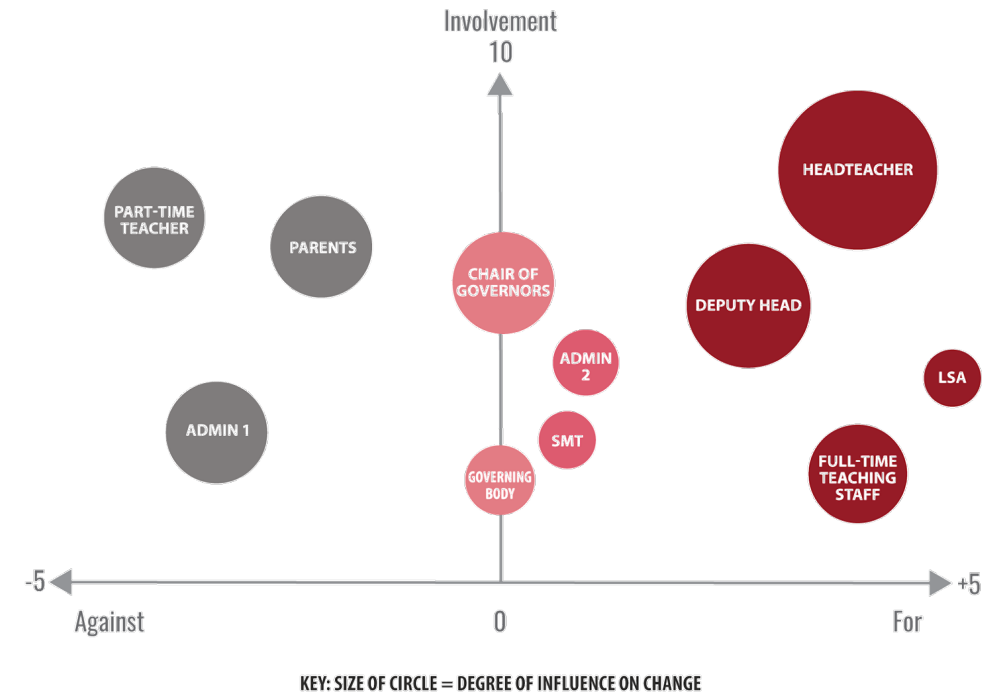


Stakeholders' map

A stakeholders' map includes the different types of players characterising the **domain of interest** and describes their relationship.

A stakeholders map usually identifies:

- **Direct or Primary users (CORE players)** as end-users, will directly affect the final results (acceptance and adoption). Their participation in activities is a requirement.
- **Indirect or Secondary users (INVOLVED players)** are those subjects that, because of their role, power, authority, responsibilities, or claims over the resources, play the role of enablers or barriers.
- **External stakeholders (INFORMED players)** are subjects that can positively contribute to supporting the solutions at a strategic level, such as the media and policy-makers.



Dolfing, H. A **Step by Step Stakeholder Mapping Guide**. 2018.

Stakeholders' management process



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Hands-on Stakeholders mapping



Design before developing.

1

- Which is your challenge?
- Which needs and opportunities do you see?
- Who will use/benefit from?
- Who will support it?

**STAKEHOLDERS
ANALYSIS & MAP**

2

- What do your customers want to reach?
- Which data and information will be provided?
- Where do they come from?

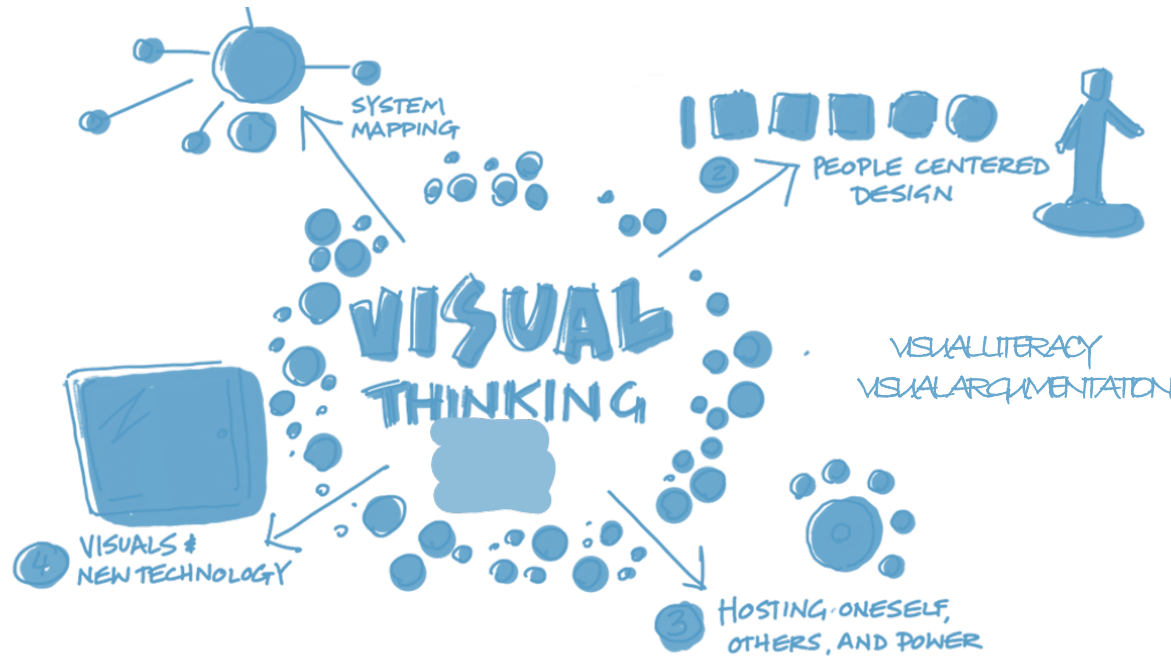
PERSONAS

ADaS Projects

PID	Title	Organisation
P1	Unsupervised topic detection in app reviews	Hype
P2	Ballpark quotation tool	IVECO
P3	AI Personas	Lavazza
P4	Automated Artwork Metadata Generation using Multimodal LLMs	LINKS
P5	DINOv3 For Damage Assessment	LINKS
P6	ENT-icipate	LINKS
P7	Multimodal Emotion Recognition	LINKS
P8	Multi-task learning for fire ignition maps	LINKS
P9	Open vocabulary Semantic segmentation	LINKS
P10	Ball and Player Tracking in Padel Matches Videos	LINKS
P11	Unsupervised energy profile clustering	LINKS
P12	RL football	PoliTo
P13	Ontology alignment	Repertorio

Show, don't tell.

It is a method for visually describing, synthesising, systematising, re-elaborating, and sharing the information, insights, models, and ideas we gain during the design process.



1) Stakeholders list

1

- **Starting from the project brief, Identify all the stakeholders**
 - Who is interested in the project
 - Who are the direct beneficiaries who will use the output?
 - Who are the indirect beneficiaries that will benefit from the outcome?
 - Who can refuse or obstruct the project/adoption?
(add 1 sticky note for every stakeholder you met during the analysis)

- **Take note of their needs and wants** (desk research or interviews)

EXAMPLES OF POSSIBLE INFORMATION

- What do they need or want? (GOALS)
- What skills do they have in the field?
- What systems do they already use to reach their goal?
- What barriers do they pose?

- **Explore the provided Datasets to search for the underlying model (mental, conceptual, operational)**
- **Integrate your domain knowledge** with desk analysis
(Reports, Research papers, Newspaper articles)

2) Stakeholders' map

2

- **Prepare your canvas and define the properties of the areas**

Direct	Core	Beneficiaries	...
Indirect	Involved	Technicians	...
exluded	Informed	Managers	...

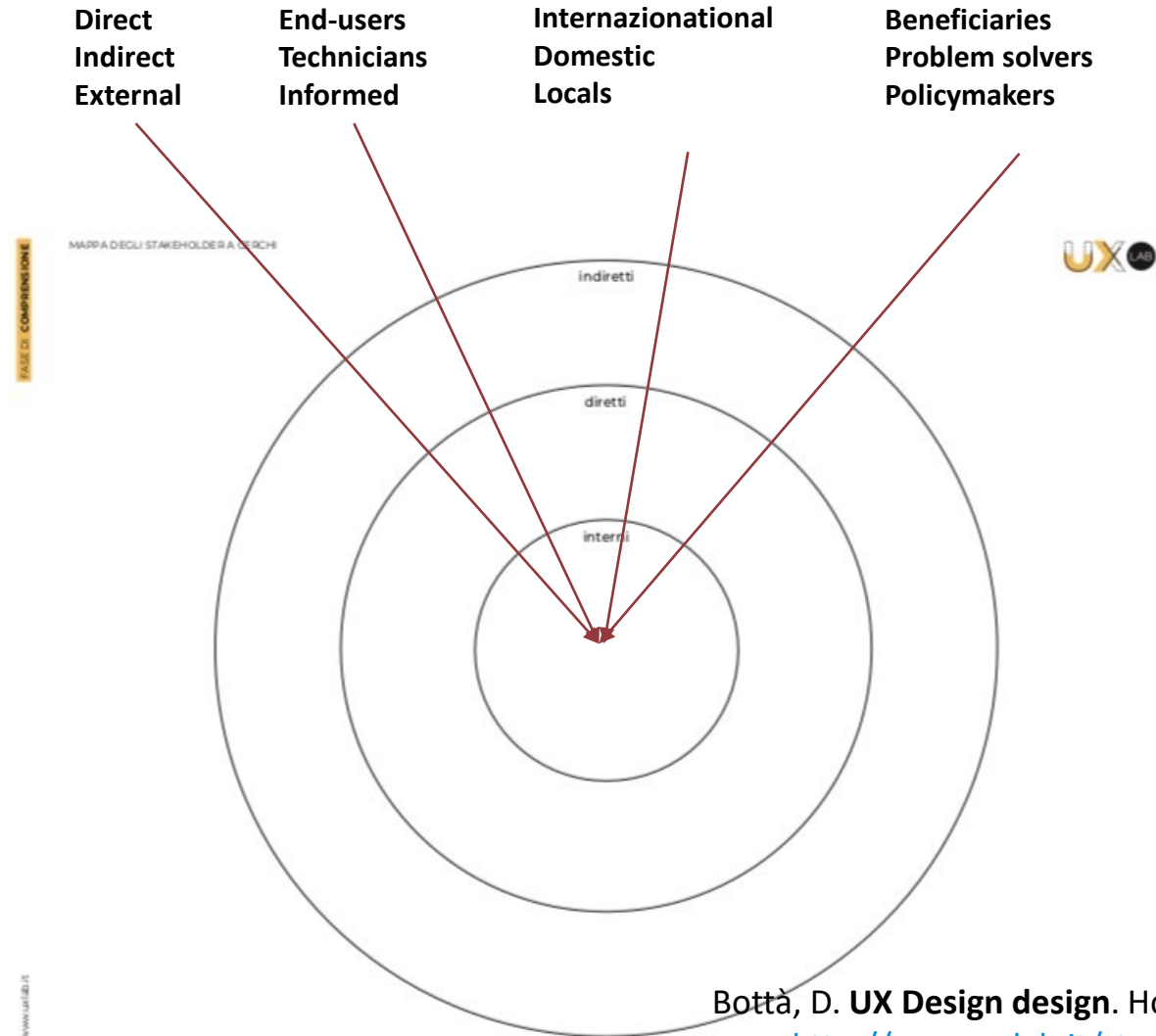
- **Move the stakeholders on the map:** place each stakeholder into the corresponding category
 - Use the space to represent commonality, connections (proximity) and size or priority (size, color)

Different visual canvases allow you to display an overview of a complex ecosystem (and then share, process, communicate on and with) of who and how you are trying to reach.

Radar chart

Useful to organize groups on one variable dimensions:

- responsibility
- influence
- proximity
- dependence, involvement
- ...

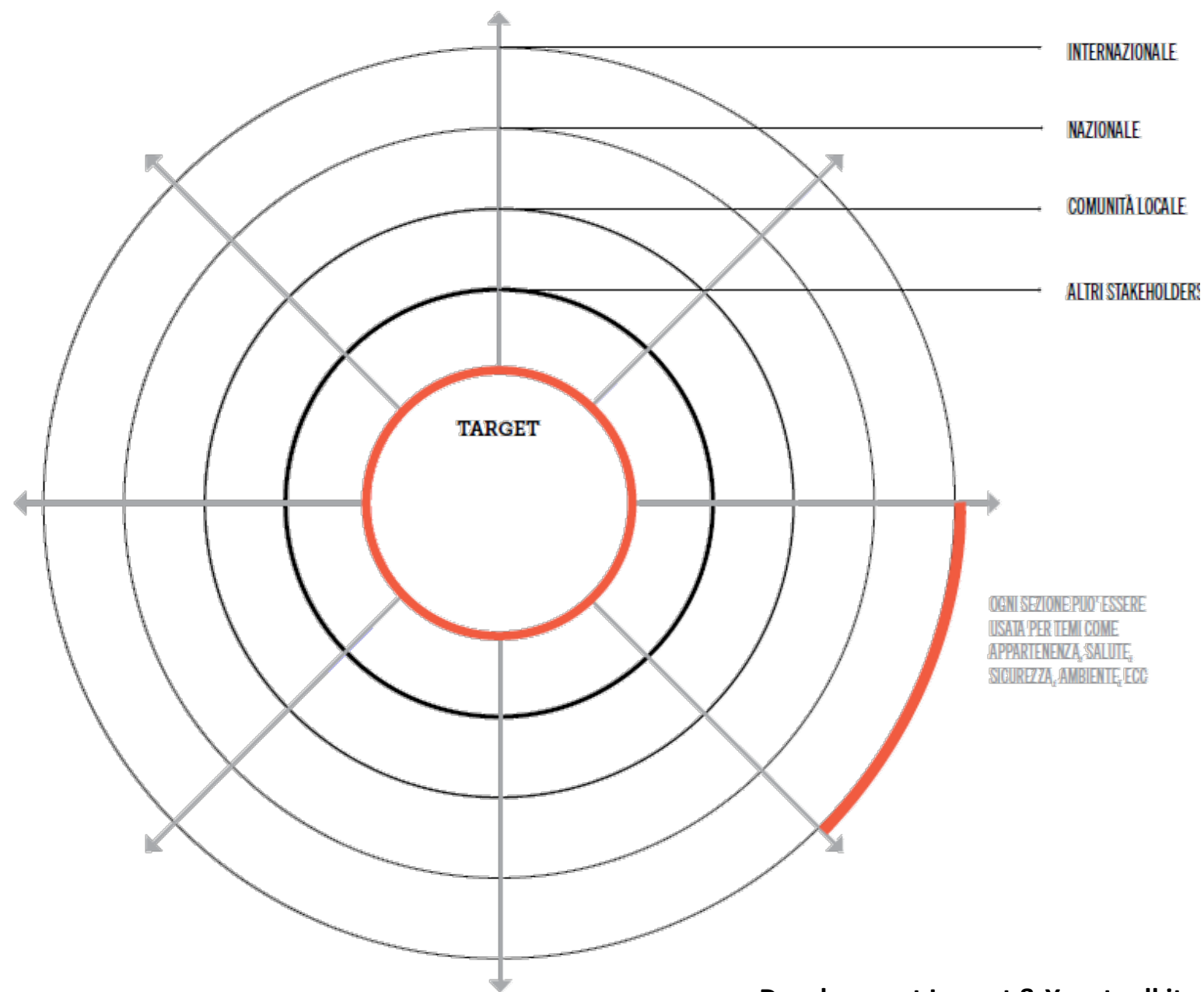


Bottà, D. **UX Design design**. Hoepli, 2019.

<http://www.uxlab.it/strumenti.html>

Radar chart

Original chart developed from
the Helsinki Design Lab .

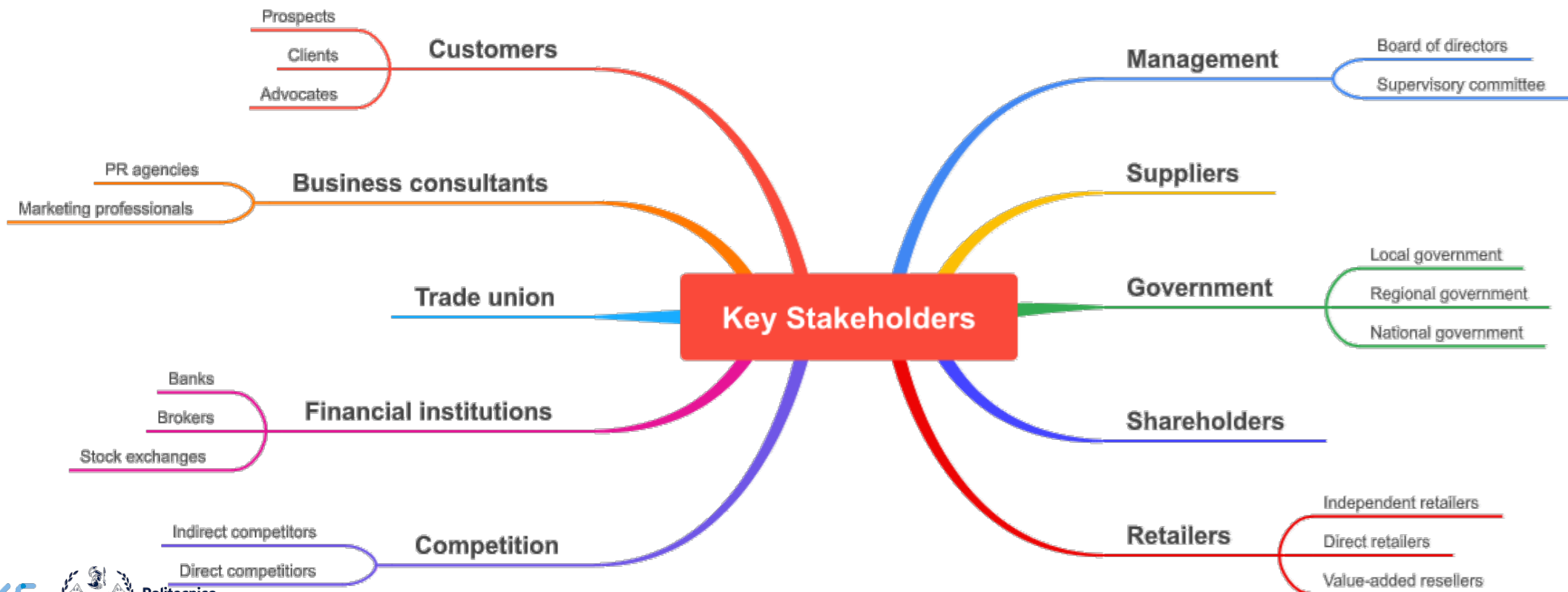


Hub & spoke

Useful to explore a key group and the other actors around it.
The visual encoding helps to describe the relationships between nodes:

- **Lines:** continuous, dashed or more marked depending on the type of relationship; **Colors:** type of target; **Size:** dimension of target.

For each branch, quantitative data that complete the picture shall be added and kept updated.



Ecosystem canvas

ECOSYSTEM CANVAS

Group:

City:

Which are the main actors of the scenario?

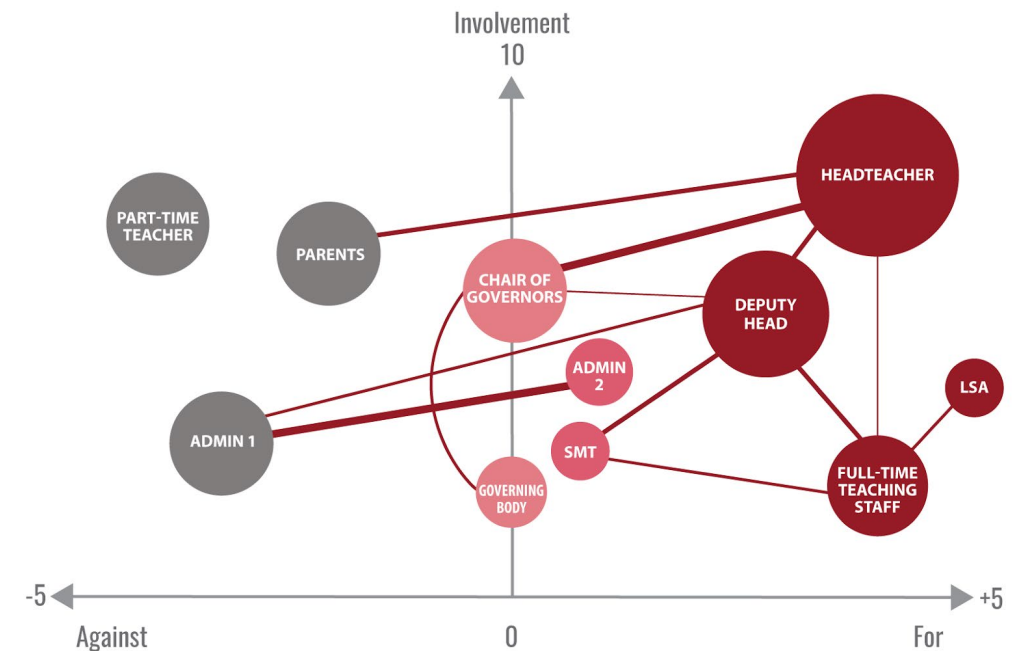
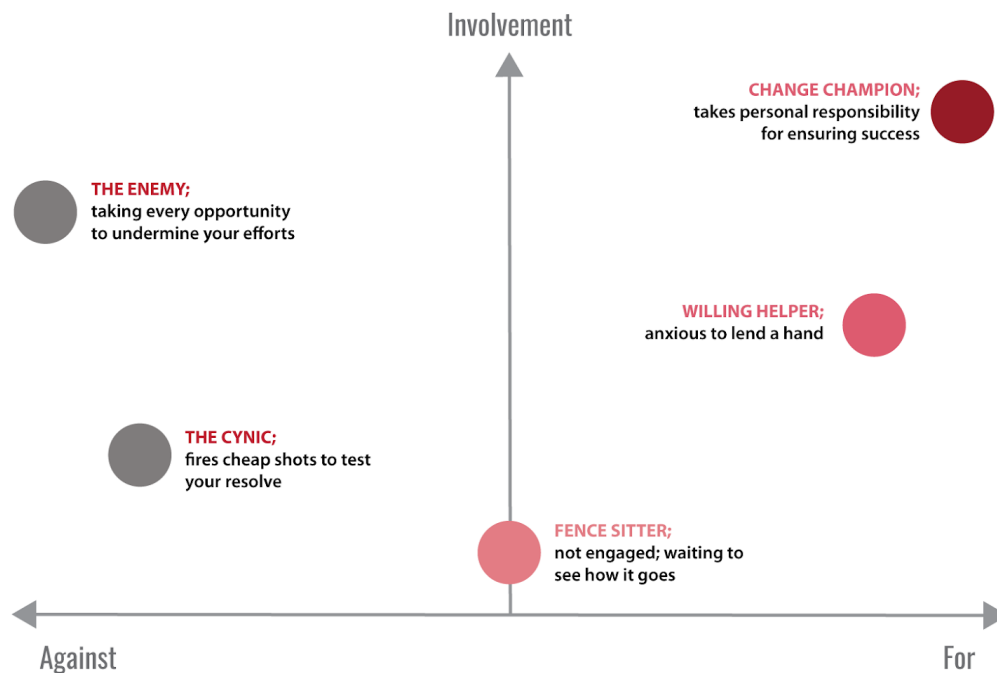
HOW TO USE IT: Identify and describe the stakeholders in 3 main roles: Peer Consumers, Peer Producers, Partners. Each role can have a specific interest in the service success, in controlling externalities and outcomes, in regulating it or in exercising rights in the service governance: they can be public actors or bodies dealing with regulation and control of service on a local basis, representatives of communities of peers and partners involved in the value creation, pre-existing institutions, associations, citizens.

Scenario



Bubble diagram

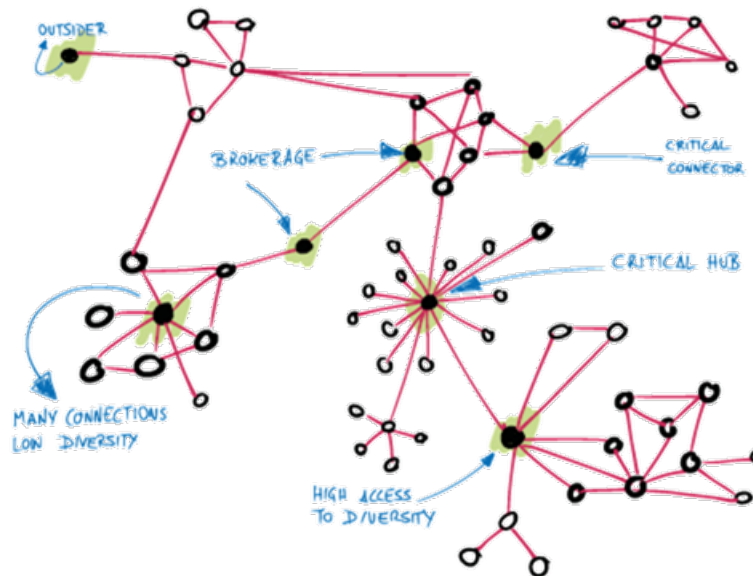
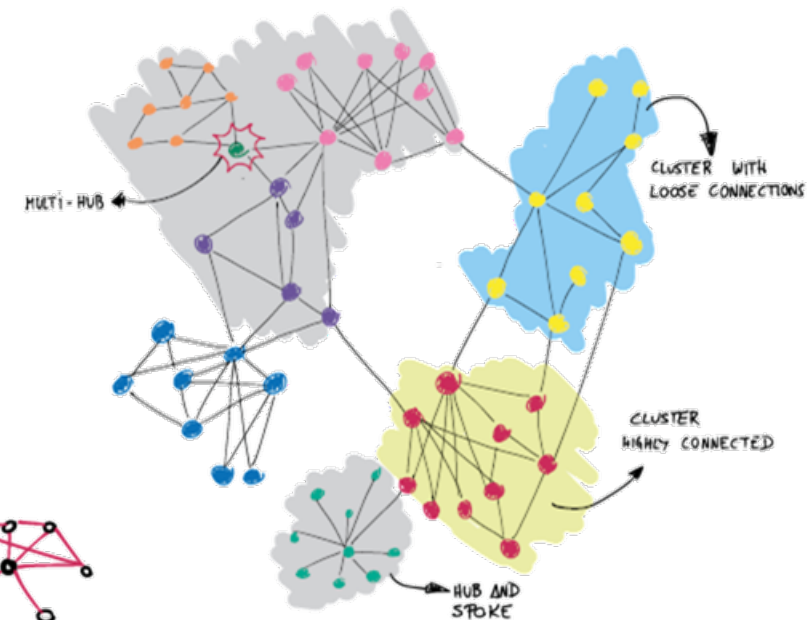
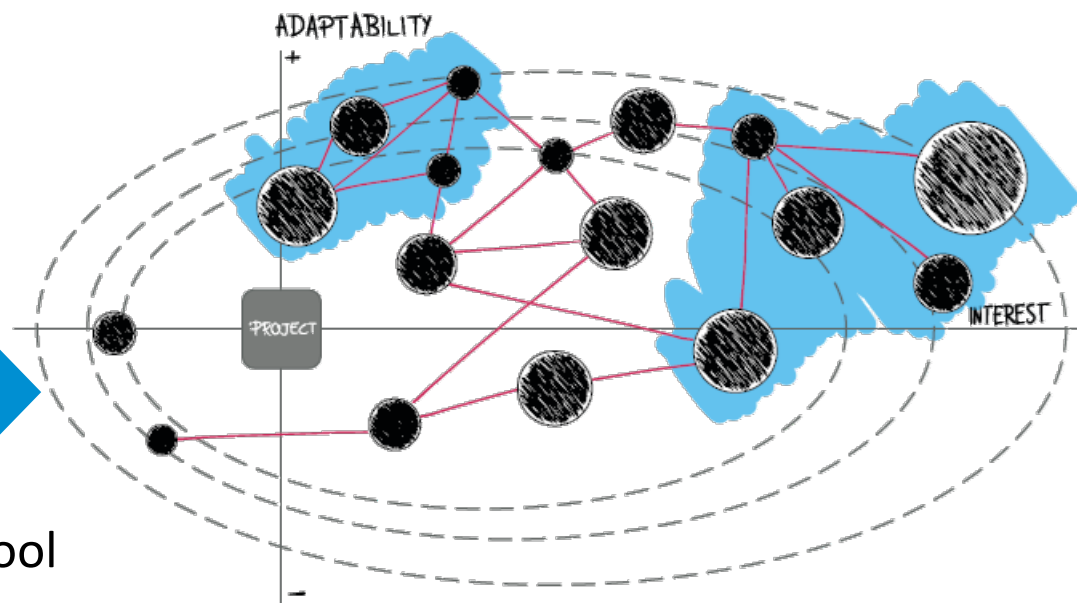
A map is a dynamic tool that should be kept up to date, to provide the project team with a synthetic view of the relations among groups and goals.



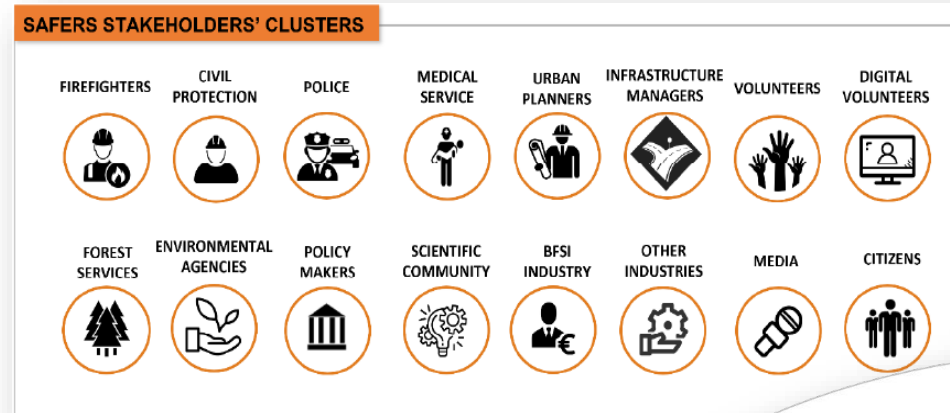
Stakeholders' universe

Visual network analysis tool focusing on the connections among actors, how they work, and how they might evolve, affecting the system we are designing.

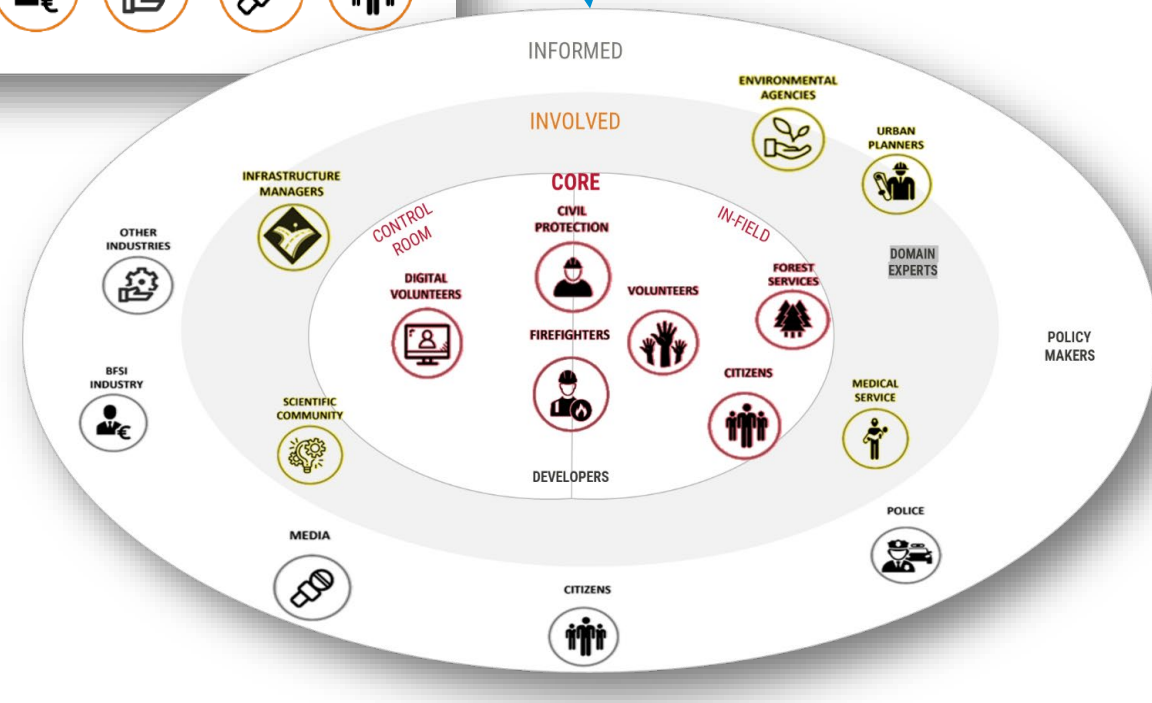
- 1) List the actors
- 2) Place them on the axis
- 3) Define connections
- 4) Define clusters



Stakeholders list



Stakeholders' map supporting the User Requirements collection And technical specifications

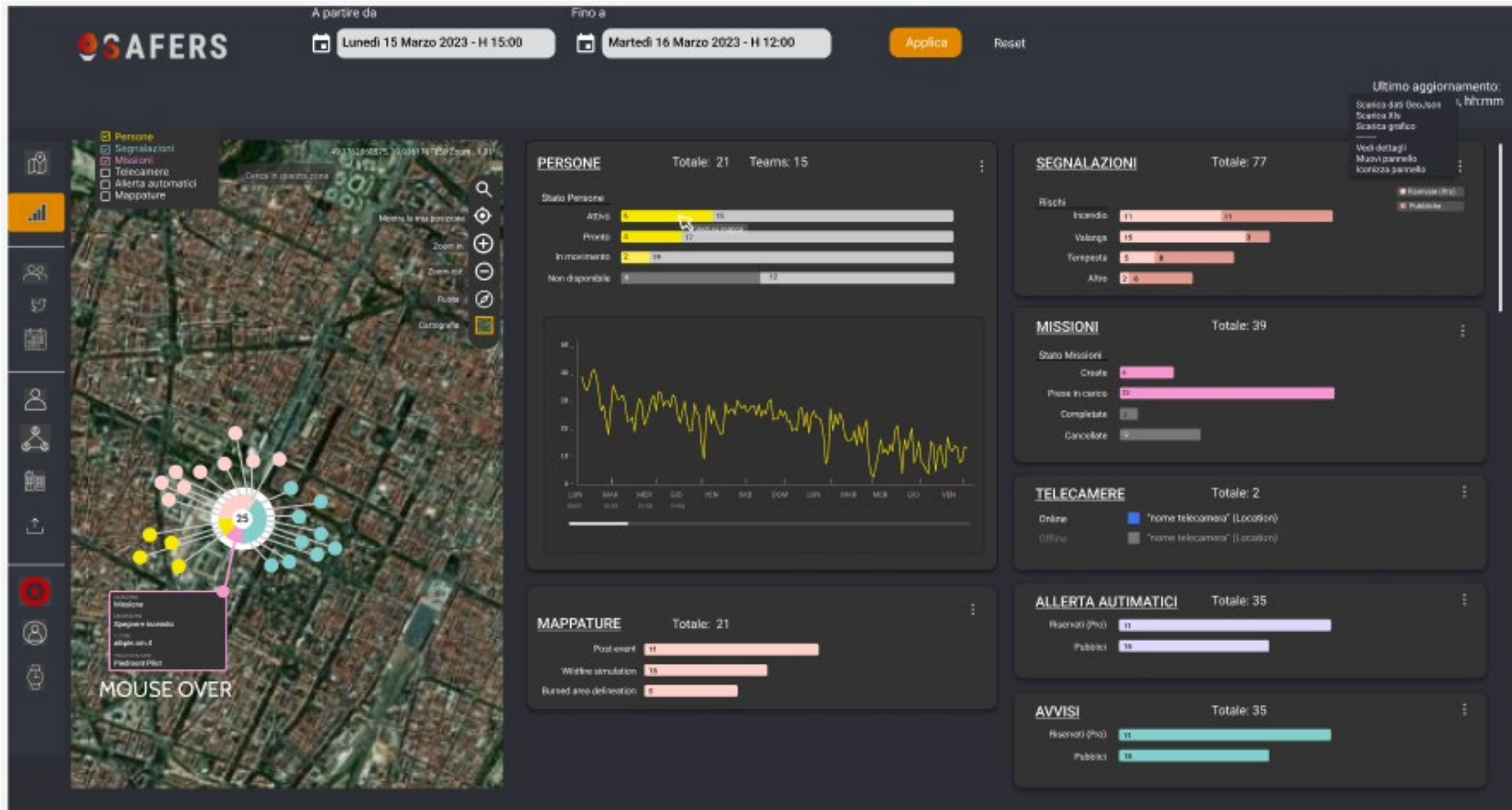


Stakeholders' table to support the Pilot design and exploitation

Stakeholder category	Role in SAFERS
Firefighters	End-users of SAFERS platform and smart services.
Civil protection	Their contribution revolves around the definition of end-user requirements, the co-design SAFERS solutions, the realization of pilots for testing and demonstrations for validation, and the provision of feedbacks. They will benefit from SAFERS during the emergency management phases: prevention and preparedness and detection and response.

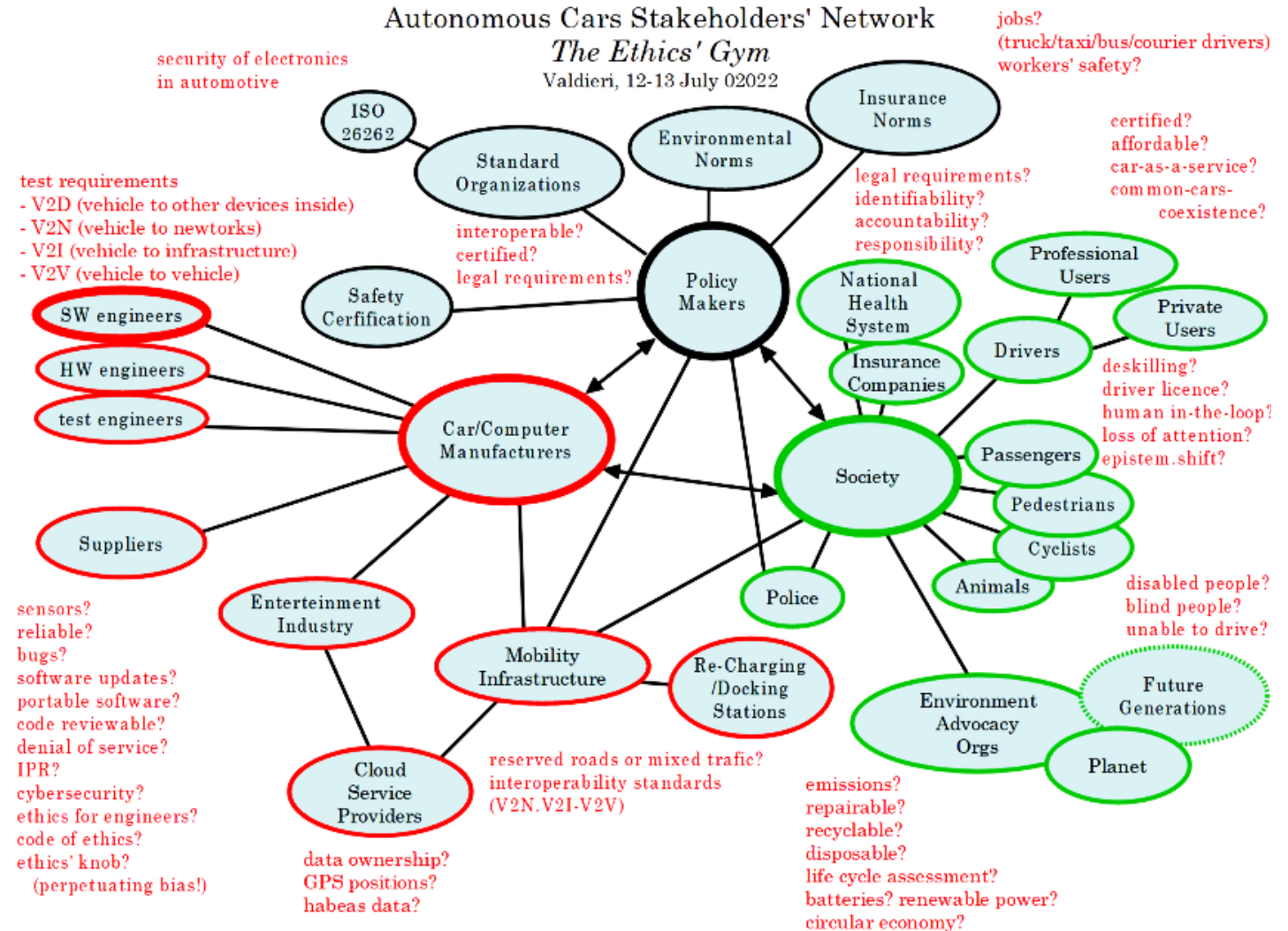


Web Dashboard
for the
consultation of
satellite data,
sensors and
reports made by
first responders
and citizens
(crowdsourcing)



Stakeholders map of the project

- Learn the *ars interrogandi*
- Learn the context
- Learn **who will use the data and system and what relationships people have with it**
- Design the network of people
- Are relations equal?
- Let **ethical issues** emerge
- And consequently, the technical ones (e.g. interoperability, inspectability)



Different visual canvases allow you to display an **overview of a complex ecosystem** (and then share, process, communicate on and with) of who and how you are trying to reach.

There are no standard maps.

You can find many different canvases and labels (ecosystem canvas, players map, ...), made available in wonderful existing toolkits.

You can also create yours for ad-hoc purposes and specific needs.

The golden rule is:

Keep it

. **simple and clear**

. **stick to data (real world)**

. **updated**

3) Stakeholders' analysis

3

- Refine the characteristics:
 - Which **roles** do they play (with reference to the specific context you're focusing on)
 - What are the **NEEDS**?
 - What are their **GOALS**?
 - **Did you forget any** stakeholder group?
 - Do you need **different groups** or labels? Refine the first list, no problem.
- **Group or connect subjects that have something in common**



Contacts

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